

Auto Scaling Group

Auto Scaling automatically **adds (scales out)** or **removes (scales in)** EC2 instances based on your application's demand.

1. Create Launch Template

A **Launch Template** in AWS is a reusable blueprint that contains all the settings needed to launch an EC2 instance.

Create launch template

Creating a launch template allows you to create a saved instance configuration that can be reused, shared and launched at a later time. Templates can have multiple versions.

Launch template name and description

Launch template name - *required*

Auto-Scaling-Template

Must be unique to this account. Max 128 chars. No spaces or special characters like '&', "'", '@'.

Template version description

Auto Scaling Group

Max 255 chars

Auto Scaling guidance [Info](#)

Select this if you intend to use this template with EC2 Auto Scaling

☒ Provide guidance to help me set up a template that I can use with EC2 Auto Scaling

► **Template tags**

► **Source template**

Summary

Software Image (AMI)

-

Virtual server type (instance type)

-

Firewall (security group)

-

Storage (volumes)

-

[Cancel](#) [Create launch template](#)

Application and OS Images (Amazon Machine Image) - required [Info](#)

An AMI contains the operating system, application server, and applications for your instance. If you don't see a suitable AMI below, use the search field or choose [Browse more AMIs](#).

Q Search our full catalog including 1000s of application and OS images

Recents **Quick Start**

Amazon Linux macOS Ubuntu Windows Red Hat SUSE Linux Debian

Amazon Machine Image (AMI)

Amazon Linux 2023 kernel-6.1 AMI

ami-0c7d68785ec07306c (64-bit (x86), uefi-preferred) / ami-078b222939550364c (64-bit (Arm), uefi)

Virtualization: hvm ENA enabled: true Root device type: ebs

Free tier eligible

Description

Amazon Linux 2023 (kernel-6.1) is a modern, general purpose Linux-based OS that comes with 5 years of long term support. It is optimized for AWS and designed to provide a secure, stable and high-performance execution environment to develop and run your cloud applications.

Amazon Linux 2023 AMI 2023.9.20251110.1 x86_64 HVM kernel-6.1

Summary

Software Image (AMI)

Amazon Linux 2023 AMI 2023.9.2...[read more](#)

ami-0c7d68785ec07306c

Virtual server type (instance type)

t3.micro

Firewall (security group)

-

Storage (volumes)

1 volume(s) - 8 GiB

[Cancel](#) [Create launch template](#)

▼ Instance type [Info](#) | [Get advice](#)

Advanced

Instance type

t3.micro

Free tier eligible

Family: t3 2 vCPU 1 GiB Memory Current generation: true
On-Demand Ubuntu Pro base pricing: 0.0143 USD per Hour On-Demand RHEL base pricing: 0.0396 USD per Hour
On-Demand SUSE base pricing: 0.0108 USD per Hour On-Demand Linux base pricing: 0.0108 USD per Hour
On-Demand Windows base pricing: 0.02 USD per Hour

☐ All generations

[Compare instance types](#)

[Additional costs apply for AMIs with pre-installed software](#)

Software Image (AMI)

Amazon Linux 2023 AMI 2023.9.2...[read more](#)
ami-0c7d68785ec07306c

Virtual server type (instance type)

t3.micro

Firewall (security group)

-

Storage (volumes)

1 volume(s) - 8 GiB

[Cancel](#)

[Create launch template](#)

▼ Key pair (login) [Info](#)

You can use a key pair to securely connect to your instance. Ensure that you have access to the selected key pair before you launch the instance.

Key pair name

Practice

[Create new key pair](#)

▼ Network settings [Info](#)

Subnet [Info](#)

Don't include in launch template

[Create new subnet](#)

When you specify a subnet, a network interface is automatically added to your template.

Availability Zone [Info](#)

Don't include in launch template

[Enable additional zones](#)

Not applicable for EC2 Auto Scaling

Firewall (security groups) [Info](#)

A security group is a set of firewall rules that control the traffic for your instance. Add rules to allow specific traffic to reach your instance.

☐ Select existing security group

☒ Create security group

Security group name - *required*

Auto-Scaling-Group

This security group will be added to all network interfaces. The name can't be edited after the security group is created. Max length is 255 characters. Valid characters: a-z, A-Z, 0-9, spaces, and _-./@#%&'()*+,-=:&[]!\$*

Description - *required* [Info](#)

Auto-Scaling-Group

VPC [Info](#)

vpc-09b6bff8ffa12c928
172.31.0.0/16

(default)

▼ Summary

Software Image (AMI)

Amazon Linux 2023 AMI 2023.9.2...[read more](#)
ami-0c7d68785ec07306c

Virtual server type (instance type)

t3.micro

Firewall (security group)

New security group

Storage (volumes)

1 volume(s) - 8 GiB

[Cancel](#)

[Create launch template](#)

aws

Search

[Alt+S]

Europe (Stockholm) Account ID: 0950-7410-7161 Sakshi Jain

EC2 > Launch templates > Create launch template

Inbound Security Group Rules

▼ Security group rule 1 (TCP, 80, 0.0.0.0/0)

Type [Info](#)

HTTP

Protocol [Info](#)

TCP

Port range [Info](#)

80

Source type [Info](#)

Anywhere

Source [Info](#)

Q Add CIDR, prefix list or security group

0.0.0.0/0 X

Description - optional [Info](#)

e.g. SSH for admin desktop

Remove

▼ Security group rule 2 (TCP, 22, 0.0.0.0/0)

Type [Info](#)

ssh

Protocol [Info](#)

TCP

Port range [Info](#)

22

Source type [Info](#)

Anywhere

Source [Info](#)

Q Add CIDR, prefix list or security group

0.0.0.0/0 X

Description - optional [Info](#)

e.g. SSH for admin desktop

Remove

▼ Security group rule 3 (TCP, 443, 0.0.0.0/0)

Type [Info](#)

HTTP

Protocol [Info](#)

TCP

Port range [Info](#)

443

Summary

▼ Security group rule 1 (TCP, 80, 0.0.0.0/0)

Type [Info](#)

HTTP

Protocol [Info](#)

TCP

Port range [Info](#)

80

Source type [Info](#)

Anywhere

Source [Info](#)

Q Add CIDR, prefix list or security group

0.0.0.0/0 X

Description - optional [Info](#)

e.g. SSH for admin desktop

Remove

▼ Security group rule 2 (TCP, 22, 0.0.0.0/0)

Type [Info](#)

ssh

Protocol [Info](#)

TCP

Port range [Info](#)

22

Source type [Info](#)

Anywhere

Source [Info](#)

Q Add CIDR, prefix list or security group

0.0.0.0/0 X

Description - optional [Info](#)

e.g. SSH for admin desktop

Remove

▼ Security group rule 3 (TCP, 443, 0.0.0.0/0)

Type [Info](#)

HTTP

Protocol [Info](#)

TCP

Port range [Info](#)

443

Summary

Software Image (AMI)

Amazon Linux 2023 AMI 2023.9.2...[read more](#)
ami-0c7d68785ec07306c

Virtual server type (instance type)

t3.micro

Firewall (security group)

New security group

Storage (volumes)

1 volume(s) - 8 GiB

[Cancel](#)

[Create launch template](#)

In Advance Setting add below script

Allow tags in metadata | Info
Don't include in launch template

User data - optional | Info
Upload a file with your user data or enter it in the field.
[Choose file](#)

```
#!/bin/bash
sudo yum install httpd -y
sudo systemctl start httpd
sudo systemctl enable httpd
sudo systemctl status httpd
echo "<h1>This is Launch Template website $HOSTNAME </h1>" > /var/www/html/index.html
```

☐ User data has already been base64 encoded

Summary
Software Image (AMI)
Amazon Linux 2023 AMI 2023.9.2...[read more](#)
ami-0c7d68785ec07306c
Virtual server type (instance type)
t3.micro
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New security group
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1 volume(s) - 8 GiB

[Cancel](#) [Create launch template](#)

EC2 > Launch templates > Create launch template

Launch Templates (1) | Info

Search

☐	Launch Template ID	Launch Template Name	Default Version	Latest Version	Create Time	Created By
☐	lt-074f78c142af20231	Auto-Scaling-Template	1	1	2025-11-13T12:56:22.000Z	arn:aws:iam::095074107161:rc

[Actions](#) [Create launch template](#)

2. Create Auto Scaling Group

EC2 > Auto Scaling groups > Create Auto Scaling group

Step 1: Choose launch template (selected)
Step 2: Choose instance launch options
Step 3 - optional: Integrate with other services
Step 4 - optional: Configure group size and scaling
Step 5 - optional: Add notifications
Step 6 - optional: Add tags
Step 7: Review

Choose launch template | Info
Specify a launch template that contains settings common to all EC2 instances that are launched by this Auto Scaling group.

Name
Auto Scaling group name
Enter a name to identify the group.
Auto Scaling group
Must be unique to this account in the current Region and no more than 255 characters.

Launch template | Info
For accounts created after May 31, 2023, the EC2 console only supports creating Auto Scaling groups with launch templates. Creating Auto Scaling groups with launch configurations is not recommended but still available via the CLI and API until December 31, 2023.

Launch template
Choose a launch template that contains the instance-level settings, such as the Amazon Machine Image (AMI), instance type, key pair, and security groups.
Auto-Scaling-Template
[Create a launch template](#)

Version
1

Select all availability zones

EC2 > Auto Scaling groups > Create Auto Scaling group

Step 1
Step 2
Step 3
Step 4
Step 5
Step 6
Step 7

Review

Network info

For most applications, you can use multiple Availability Zones and let EC2 Auto Scaling balance your instances across the zones. The default VPC and default subnets are suitable for getting started quickly.

Q |

☒ eun1-az1 (eu-north-1a) | subnet-0791f94763c174151
172.31.16.0/20 Default

☒ eun1-az2 (eu-north-1b) | subnet-03c99dad9cfe50a6d
172.31.32.0/20 Default

☒ eun1-az3 (eu-north-1c) | subnet-04712f4123f4a8913
172.31.0.0/20 Default

Select Availability Zones and subnets

eun1-az1 (eu-north-1a) | subnet-0791f94763c174151
172.31.16.0/20 Default

eun1-az2 (eu-north-1b) | subnet-03c99dad9cfe50a6d
172.31.32.0/20 Default

eun1-az3 (eu-north-1c) | subnet-04712f4123f4a8913
172.31.0.0/20 Default

Create a subnet

Availability Zone distribution - new

Auto Scaling automatically balances instances across Availability Zones. If launch failures occur in a zone, select a strategy.

☒ **Balanced best effort**
If launches fail in one Availability Zone, Auto Scaling will attempt to launch in another healthy

☐ **Balanced only**
If launches fail in one Availability Zone, Auto Scaling will continue to attempt to launch in the

New load balancer

EC2 > Auto Scaling groups > Create Auto Scaling group

Step 1
Step 2
Step 3
Step 4
Step 5
Step 6
Step 7

Choose launch template
Choose instance launch options
Integrate with other services
Configure group size and scaling
Add notifications
Add tags
Review

Integrate with other services - optional info

Use a load balancer to distribute network traffic across multiple servers. Enable service-to-service communications with VPC Lattice. Shift resources away from impaired Availability Zones with zonal shift. You can also customize health check replacements and monitoring.

Load balancing info

Use the options below to attach your Auto Scaling group to an existing load balancer, or to a new load balancer that you define.

☐ No load balancer
Traffic to your Auto Scaling group will not be fronted by a load balancer.

☐ Attach to an existing load balancer
Choose from your existing load balancers.

☒ **Attach to a new load balancer**
Quickly create a basic load balancer to attach to your Auto Scaling group.

Attach to a new load balancer

Load balancer type
Choose from the load balancer types offered below. Type selection cannot be changed after the load balancer is created. If you need a different type of load balancer than those offered here, visit the [Load Balancing console](#).

☒ **Application Load Balancer**
HTTP, HTTPS

☐ Network Load Balancer
TCP, UDP, TLS

Load balancer name
Name cannot be changed after the load balancer is created.

Auto Scaling group-1

Listeners and routing

If you require secure listeners, or multiple listeners, you can configure them from the [Load Balancing console](#) after your load balancer is created.

Protocol

HTTP

Port

80

Default routing (forward to)

Create a target group

New target group name
An instance target group with default settings will be created.

AutoScalinggroup

Tags - optional

Consider adding tags to your load balancer. Tags enable you to categorize your AWS resources so you can more easily manage them.

Add tag

50 remaining

During an Availability Zone impairment, target instance launches towards other healthy Availability Zones.

☐ Enable zonal shift
New instance launches will be retargeted towards healthy Availability Zones until the zonal shift is canceled.

Health checks
Health checks increase availability by replacing unhealthy instances. When you use multiple health checks, all are evaluated, and if at least one fails, instance replacement occurs.

EC2 health checks
[Always enabled](#)

Additional health check types - optional [Info](#)

☐ Turn on Elastic Load Balancing health checks **Recommended**
Elastic Load Balancing monitors whether instances are available to handle requests. When it reports an unhealthy instance, EC2 Auto Scaling can replace it on its next periodic check.

☐ Turn on VPC Lattice health checks
VPC Lattice can monitor whether instances are available to handle requests. If it considers a target as failed a health check, EC2 Auto Scaling replaces it after its next periodic check.

☐ Turn on Amazon EBS health checks
EBS monitors whether an instance's root volume or attached volume stalls. When it reports an unhealthy volume, EC2 Auto Scaling can replace the instance on its next periodic health check.

Health check grace period [Info](#)
This time period delays the first health check until your instances finish initializing. It doesn't prevent an instance from terminating when placed into a non-running state.

10 seconds

[Cancel](#) [Skip to review](#) [Previous](#) [Next](#)

Desired capacity: The number of EC2 instances you want to run **right now**.

Min Desired capacity: The lowest number of EC2 instances that should always be running — even when there is no traffic.

Max desired capacity: This is the maximum number of EC2 instances Auto Scaling is allowed to create.

Step 1 Choose launch template
Step 2 Choose instance launch options
Step 3 - optional Integrate with other services
Step 4 - optional **Configure group size and scaling**
Step 5 - optional Add notifications
Step 6 - optional Add tags
Step 7 Review

Configure group size and scaling - optional [Info](#)
Define your group's desired capacity and scaling limits. You can optionally add automatic scaling to adjust the size of your group.

Group size [Info](#)
Set the initial size of the Auto Scaling group. After creating the group, you can change its size to meet demand, either manually or by using automatic scaling.

Desired capacity type
Choose the unit of measurement for the desired capacity value. vCPUs and Memory(GiB) are only supported for mixed instances groups configured with a set of instance attributes.

Units (number of instances)

Desired capacity
Specify your group size.

2

Scaling [Info](#)
You can resize your Auto Scaling group manually or automatically to meet changes in demand.

Scaling limits
Set limits on how much your desired capacity can be increased or decreased.

Min desired capacity **Max desired capacity**

1 4

Equal or less than desired capacity Equal or greater than desired capacity

Instance warmup: Instance Warmup is the time Auto Scaling waits **after launching a new EC2 instance** before it starts using that instance's metrics (like CPU) for scaling decisions.

EC2 > Auto Scaling groups > Create Auto Scaling group

Select whether you want Auto Scaling to launch instances into an existing Capacity Reservation or Capacity Reservation resource group.

- ☒ **Default**
Auto Scaling uses the Capacity Reservation preference from your launch template.
- ☐ **None**
Instances will not be launched into a Capacity Reservation.
- ☐ **Capacity Reservations only**
Instances will only be launched into a Capacity Reservation. If capacity isn't available, the instances fail to launch.
- ☐ **Capacity Reservations first**
Instances will attempt to launch into a Capacity Reservation first. If capacity isn't available, instances will run in On-Demand capacity.

Additional settings

Instance scale-in protection
If protect from scale in is enabled, newly launched instances will be protected from scale in by default.

☐ Enable instance scale-in protection

Monitoring [Info](#)
☐ Enable group metrics collection within CloudWatch

Default instance warmup [Info](#)
The amount of time that CloudWatch metrics for new instances do not contribute to the group's aggregated instance metrics, as their usage data is not reliable yet.

☒ Enable default instance warmup

60 seconds

Cancel Skip to review Previous Next

Auto Scaling groups (1) [Info](#)

Last updated less than a minute ago

[Launch configurations](#) [Launch templates](#) [Actions](#) [Create Auto Scaling group](#)

Search your Auto Scaling groups

☐	Name	Launch template/configuration	Instances	Status	Desired capacity	Min	Max	Availability Zones	Created
☐	Auto Scaling group	Auto-Scaling-Template Version Default	0	Updating capacity...	2	1	4	3 Availability Zones	Thu No...

As we set Desired Capacity is 2 so, we can see 2 instances are running.

EC2 > Instances

Instances (2) [Info](#)

Find Instance by attribute or tag (case-sensitive)

All states

☐	Name	Instance ID	Instance state	Instance type	Status check	Alarm status	Availability Zone	Public IPv4 DNS
☐		i-0cf0b91d3b100fc6b	Running	t3.micro	3/3 checks passed	View alarms	eu-north-1a	ec2-16-16-202-41.eu
☐		i-0c279f21824f6c1f2	Running	t3.micro	3/3 checks passed	View alarms	eu-north-1c	ec2-16-16-156-105.e



This is Launch Template website ip-172-31-28-53.eu-north-1.compute.internal

After Terminating Instance, it creates 2 Instance again because as we set instances warmup is **60 Sec**.

Successfully initiated termination (deletion) of i-0cf0b91d3b100fc6b, i-0c279f21824f6c1f2

Instances (4) Info

Find Instance by attribute or tag (case-sensitive)

	Name	Instance ID	Instance state	Instance type	Status check	Alarm status	Availability Zone	Public IPv4 DNS
☐		i-0db62ca301d74a381	Running	t3.micro	3/3 checks passed	View alarms +	eu-north-1a	ec2-16-171-113-226
☐		i-0cf0b91d3b100fc6b	Terminated	t3.micro	-	View alarms +	eu-north-1a	-
☐		i-0c279f21824f6c1f2	Terminated	t3.micro	-	View alarms +	eu-north-1c	-
☐		i-006601e60826220a4	Running	t3.micro	initializing	View alarms +	eu-north-1c	ec2-13-50-242-191.e